

Overview of data lakehouse architecture

As a data cloud professional, you'll work with a variety of different data sources, including structured, unstructured, and semi-structured data. Each data source is stored in different ways. So, creating an architecture to accommodate this data can help your organization use it to draw insights and make decisions. Understanding how and why data is stored will also help you design, implement, and maintain cloud data architectures, and analyze data from the cloud. In this reading, you'll learn more about data lakes, and how data lakes and cloud tools make data lakehouses possible.

By the lake

So far, you've learned that a data lake is a storage system that stores large amounts of raw data in its original format until it's needed. Like the traditional data warehouse, a data lake combines many different sources and stores them in one place. While data warehouses are hierarchical, data lakes are flat. In a data lake, data has been tagged to make it identifiable, but not organized. It's fluid, which is why it's called a data lake. Data lakes also don't require the data to be transformed before storage, so they're useful if your system is ingesting a lot of different data types.

Data lakes also have challenges. For example, they require a lot of processing to make data usable from storage, which requires an investment in resources. It's also harder to ensure data quality since data isn't cleaned immediately. This makes it difficult to ensure organizational standards are being met, and that the data being ingested is usable. Also, querying data lakes is less efficient because of the way data is organized, making it more challenging to locate requested data.

This is where data lakehouse architecture comes in. Lakehouse architecture makes the most of traditional data warehouses and the flexibility of data lakes.

Components of data lakehouse architecture

A data lakehouse is a hybrid data architecture that combines the features of a data lake with those of a data warehouse. It plays on the strengths of both data warehouses and data lakes to improve data management capabilities, seamlessly integrate with analytics tools, promote scalability and flexibility, and efficiently handle various workloads. Lakehouse architecture has five layers: ingestion, storage, metadata, API, and consumption. Review the chart to learn how each of these layers has an important role in the overall functionality of the lakehouse:

Layer	Functions
Ingestion	Retrieves data from various sources and transports it to the storage layer
Storage	Stores raw data so that users and tools have direct access to data on an as-needed basis
Metadata	Catalogs metadata about stored data to enable consistent data governance and schema management
API	Organizes APIs, analytics, and processing tools to easily run jobs
Consumption	Exists as the top layer, which can access storage, metadata, and API layers to perform analytics jobs

Key takeaways

The cloud has created so many new and innovative solutions for storing and accessing data. As organizations use the cloud more in their data systems, more solutions are being developed. Data lakehouse architecture is just one example of how to innovate on storage solutions. As a cloud data analyst, you'll have opportunities to explore cloud data architectures, and find solutions for your organization.